进度汇报

2020.4.16

胡敬玉

L. Yu et al., “Automatic 3D Cardiovascular MR Segmentation with Densely-Connected Volumetric ConvNets,” in MICCAI, 2017, pp. 287–295.

Motivation

ResNet → DenseNet

Shortcut

- **Feature reuse**
- **Supervised by loss function through short connections**

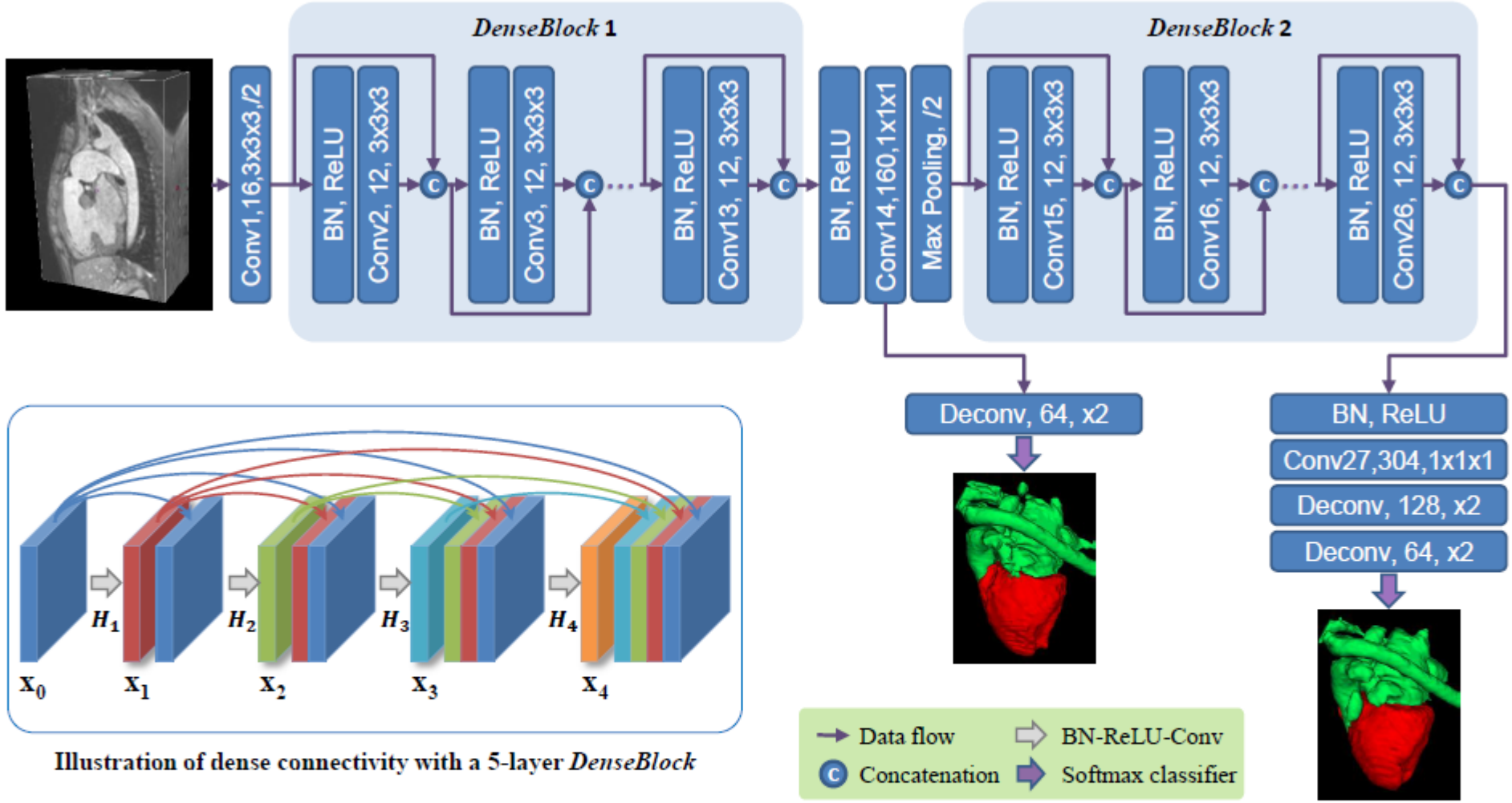
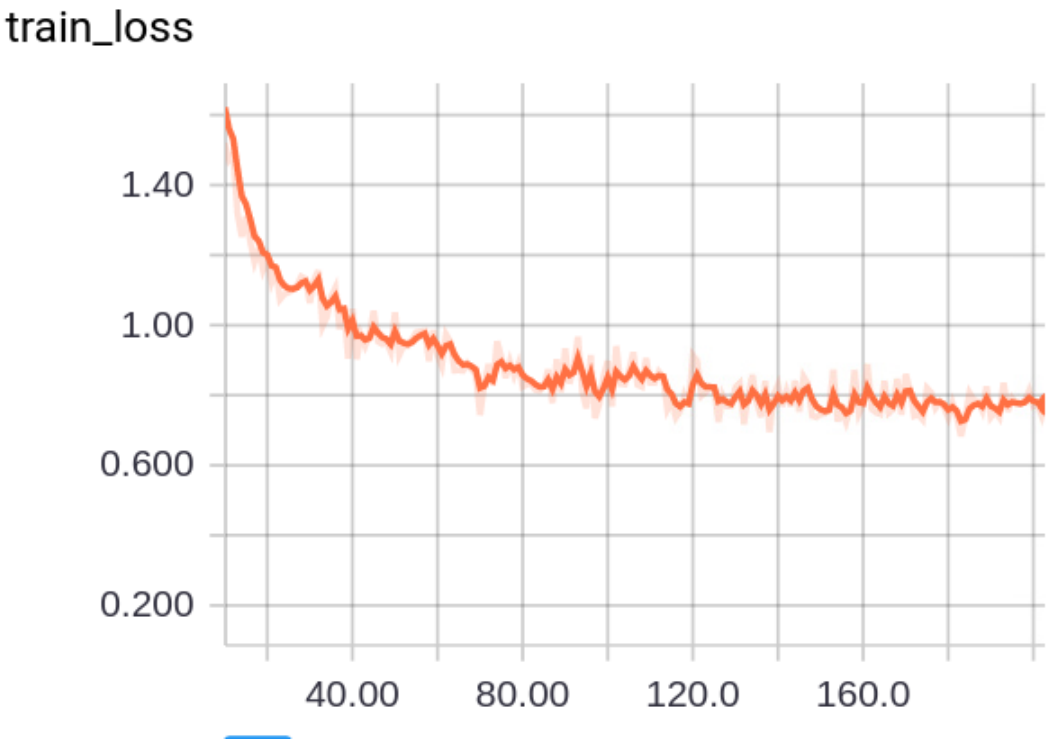


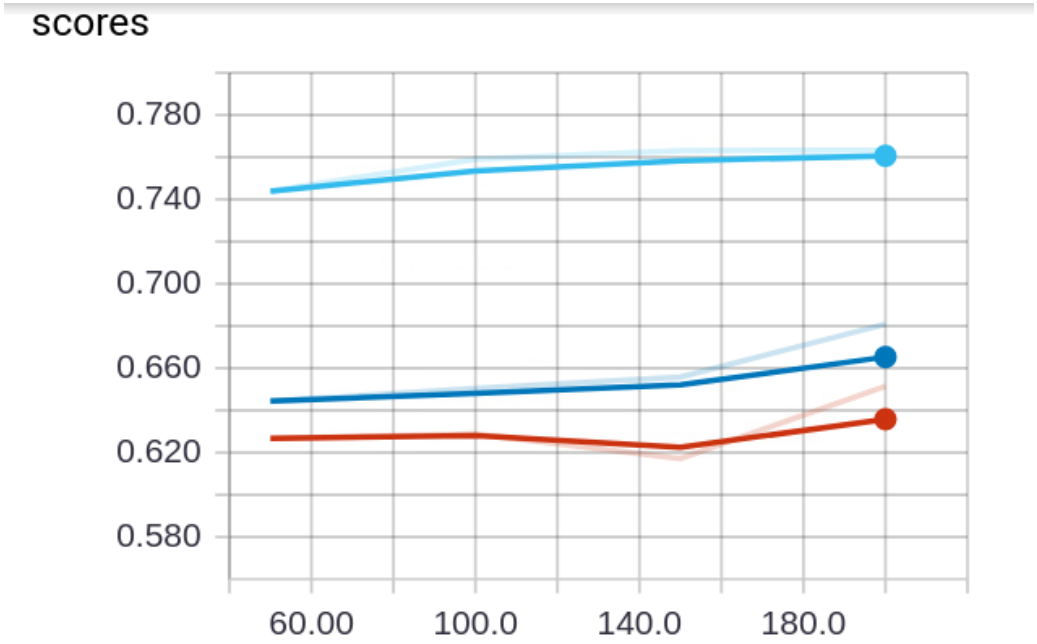
Fig. 1: The architecture of our DenseVoxNet. It consists of two *DenseBlocks* and all operations are implemented in a 3D manner. The green and red color denotes the output of blood pool and myocardium. The graph in left bottom illustrates the dense connectivity scheme taking a 5-layer *DenseBlock* as an example.

实验 DenseVoxNet

Optimizer: SGD



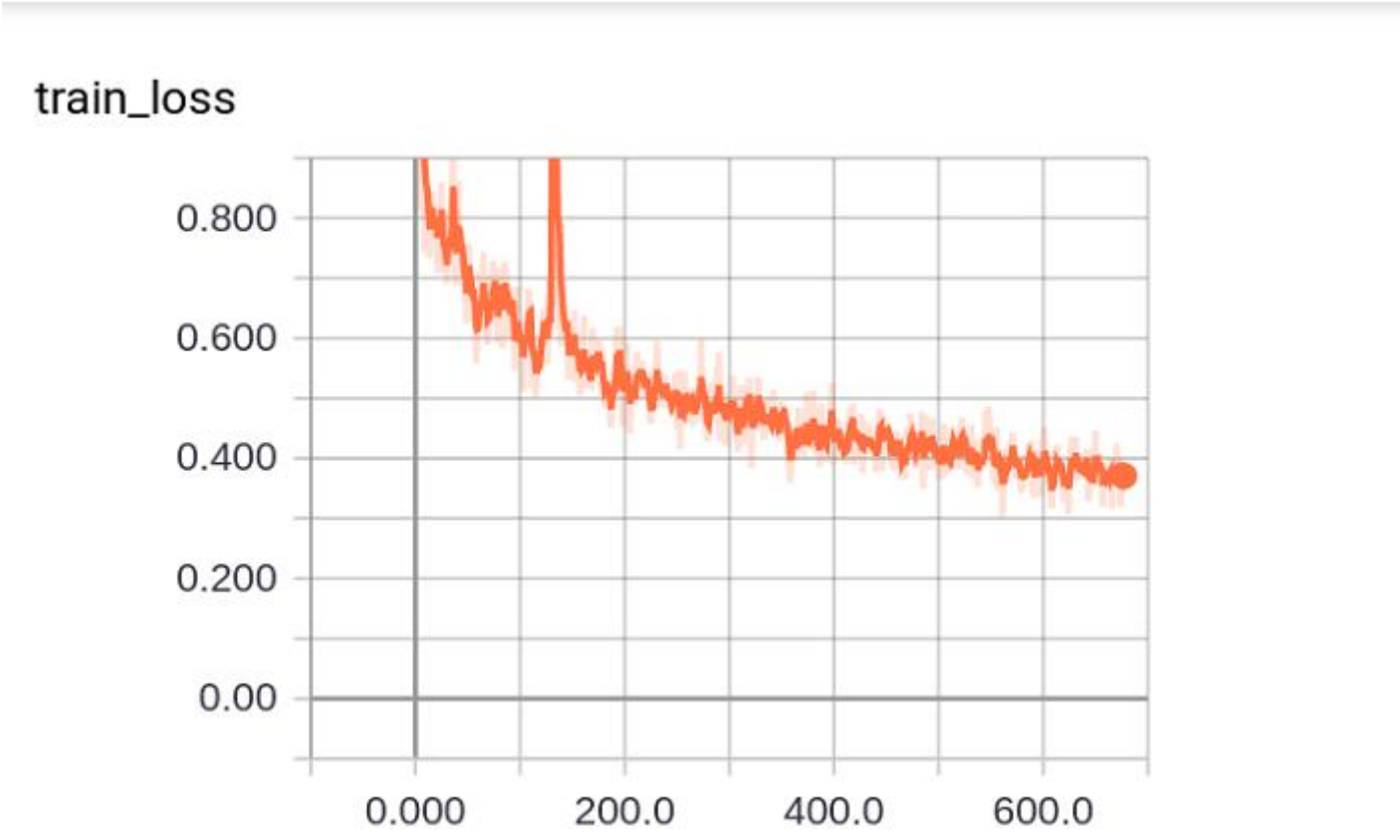
Dice



Complete
Core
Enhancing

实验 DenseVoxNet

Optimizer: Adam



实验 DenseVoxNet

Method	Dice Coefficient		
	Complete	Core	Enhancing
DenseVoxNet	0.8533	0.8352	0.7829
Chen et al.[21]	0.72	0.83	0.81
Dolz et al.[20]	0.8389	0.7968	0.7987
Proposed	0.8945	0.8483	0.8303
Re-implement(Pytorch)	0.8748	**0.8503**	0.8082

Thank You!